



MPEG-G Microbiome Classification Challenge

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Our Hackathon

- Explore the intersection of AI, microbiome science, and health.
- Work with real-world multi-omic data: microbiome, cytokines, and clinical events.
- Learn how genomics and immune profiling data can be compressed, analyzed, and modeled using modern ML techniques.
- Collaborate with experts and peers on pressing challenges in personalized medicine.
- Develop tools and insights that advance integrative, scalable bioinformatics



What is the Microbiome

- The microbiome refers to the community of microorganisms (bacteria, viruses, fungi, archaea) living in a particular environment.
- In humans, key microbiome sites include: gut, skin, oral cavity, nasal passages, and urogenital tract.
- Important Functions: immune modulation, nutrient synthesis, pathogen defense.
- Dysbiosis (imbalance) is linked to diseases: gastrointestinal, obesity, cancer, neurodegeneration.



The Human Microbiome

Current research suggests that the number of microbial cells in the human body is roughly comparable to the number of human cells, rather than the previously believed 10:1 ratio

Estimates for humans:

- Human cells: Approximately 30 trillion.
- Bacterial cells: Approximately 38 trillion.

This gives a ratio of about 1.3 bacterial cells to every one human cell. While the cell counts are similar, the bacterial cells are much smaller, accounting for a relatively small portion of the body's total mass (around 7 ounces). Microbiome catalogs usually have ~100M genomes.

Important to note:

- The human microbiome also includes other microorganisms like fungi, viruses, and archaea, though bacteria make up the largest proportion.
- The gut microbiome is the most densely populated area, containing thousands of different bacterial species and strains.
- Changes in the microbiome composition can affect health and disease.



Stanford Longitudinal Microbiome Study

Key points

- **Individualized Microbiomes:**

A 6-year study of 86 participants across gut, skin, nose, and mouth microbiomes found that the most persistent bacteria were unique to each person, shaped by genetics, diet, and immunity — like a biological fingerprint. (Snyder et al., Stanford / NIH Human Microbiome Project)

- **Dynamic & Health-Linked:**

People with Type 2 diabetes showed a less stable and diverse microbiome, possibly due to changes in blood nutrients affecting microbial growth.

- **Massive Scope & Longitudinal Insight:**

Over 5,400 samples and 118 million data points were analyzed. The microbiome was tracked alongside infections, vaccinations, and antibiotics, revealing its dynamic nature and connection to chronic disease.

- **Key finding:**

Long-term, multi-site monitoring reveals the microbiome is a fluid, systemic ecosystem — and may hold the key to personalized health insights.

<https://med.stanford.edu/news/all-news/2024/03/personal-microbiome.html#:~:text=Scientists%20have%20recently%20gained%20an, someone's%20microbiome%20can%20alleviate%20disease.>



Let's Dive into the Data

1. Microbiome Sequencing

- **Samples:** 5,432 microbiome samples collected quarterly, plus additional samples during events (infection, vaccination, antibiotics) .
- **Sites:** Four body sites—stool (gut), oral (mouth), nasal (nose), and skin.
- **Analysis:** 16S rRNA gene sequencing to determine bacterial community composition, stability, and diversity over time.

2. Cytokines (Immune Profiling)

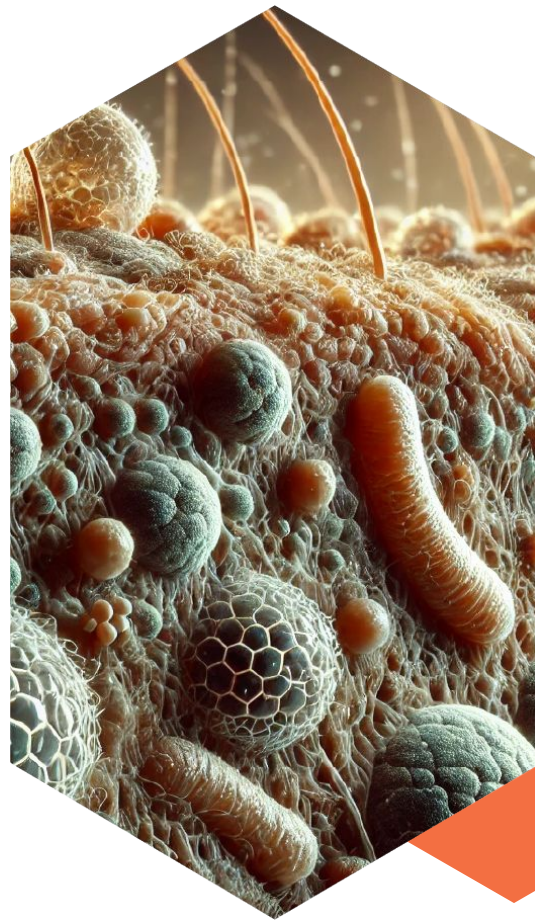
- **Measurements:** Longitudinal circulating cytokine levels, captured via multiplex assays (e.g., Luminex) and analyzed alongside microbiome dynamics [academic.oup.com+7pubmed.ncbi.nlm.nih.gov+7pmc.ncbi.nlm.nih.gov+7](https://academic.oup.com/7pubmed.ncbi.nlm.nih.gov/7pmc.ncbi.nlm.nih.gov/7)
- **Analyses:** Mixed-effects models used to identify systematic correlations between microbiome taxa and cytokine levels.

3. Health & Clinical Metadata

- **Events Recorded:** Infection episodes, vaccinations, antibiotic treatments.
- **Longitudinal Health Monitoring:** Data collected alongside microbiome samples to link physiological changes with microbial dynamics.
- **Disease Focus:** Specific attention to individuals with insulin resistance or Type 2 diabetes, where microbiome stability and diversity were perturbed

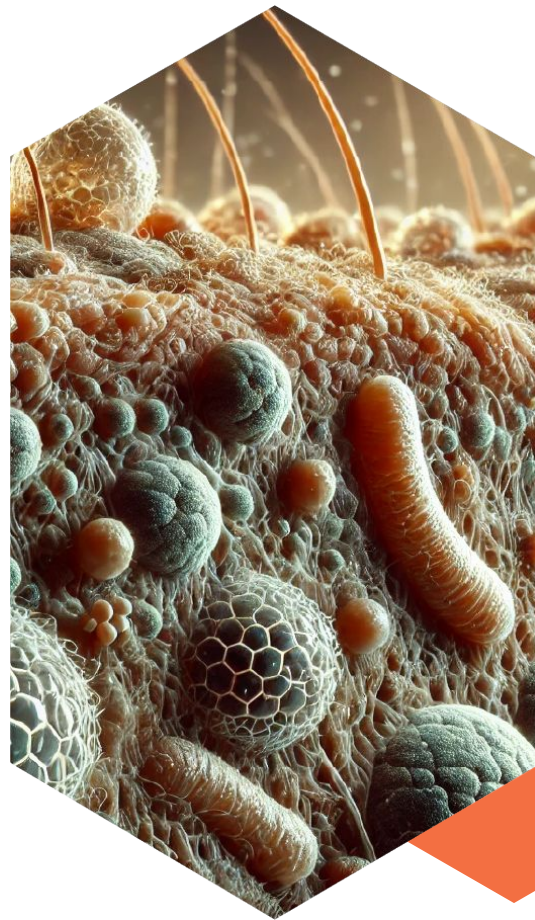
What is 16S rRNA Sequencing?

- Targets the 16S ribosomal RNA gene — conserved in bacteria but with hypervariable regions.
- Enables identification of bacterial taxa in complex samples.
- Steps: DNA extraction → PCR of 16S gene → Sequencing → Denoising/Clustering → Taxonomic classification.
- Output: OTU/ASV tables (features), diversity indices, phylogenetic trees.



16S vs. Shotgun Metagenomics

- 16S Sequencing (our data):
 - Focused on bacterial/archaeal ID
 - Lower cost, less data, taxonomic only
- Shotgun Metagenomics:
 - All DNA sequenced (bacteria, fungi, viruses)
 - Functional + taxonomic profiles
 - Higher resolution but more complex



Correlation and Causation in the Microbiome Research

	Correlation	Causation
Difficulty	Easy to design	Hard to design
Techniques	Observational study, Quantitative comparison	Interventional study, Experiments with perturbations
Study Design	Case-control study	Control-treatment study
Advantage	Large Scale, hypothesis generating	Better controlled experiments, hypothesis testing
Disadvantage	Higher false-positive rate	Difficult to have perfect control
Example	Large scale microbiome project, MetaHIT, HMP, EMP, American Gut, 500 HFGP...	Antibiotics treatment Fecal Microbiome Transplantation
Typical Findings	Correlation between microbiome composition (dysbiosis) with certain phenotype	Eliminate the microbiome could change the phenotype, or phenotype transfer with FMT

Taxonomic analysis

Microbiome taxonomy uses a hierarchical system to classify microorganisms, with levels ranging from broad domains to specific species. These levels include Domain, Kingdom, Phylum, Class, Order, Family, Genus, and Species. Understanding these levels is crucial for analyzing and interpreting microbiome data.

Microbiome Taxonomy Levels:

- [Domain](#): The highest level, grouping organisms into three domains: Bacteria, Archaea, and Eukarya.
- [Kingdom](#): A traditional level, often less emphasized in microbiome studies due to the dominance of bacteria and archaea.
- [Phylum](#): A major division within a domain, like Firmicutes and Bacteroidetes in the human gut.
- [Class](#): A subdivision of a phylum, such as Clostridia within Firmicutes.
- [Order](#): Further division within a class, like Clostridiales within Clostridia.
- [Family](#): A group of related genera.
- [Genus](#): A group of closely related species.
- [Species](#): The most specific level, representing a group of organisms that can interbreed.

ASV Vs OTU

Amplicon Sequence Variant (ASV) - the dots

An ASV is a precisely resolved DNA sequence variant inferred from amplicon data (like 16S rRNA sequencing). Tools like DADA2 or Deblur denoise raw reads to distinguish even single-nucleotide differences, providing higher resolution than OTUs.

>>ASVs are exact sequence variants, reproducible across datasets, and more biologically meaningful for tracking fine-scale microbial changes.

Operational Taxonomic Units (OTUs) - the clusters

An OTU is a cluster of DNA sequences that are grouped together based on similarity, typically at 97% identity. It serves as a proxy for a microbial "species" in early microbiome studies. OTUs help simplify analysis by grouping closely related sequences into a single unit, but they can lump together biologically distinct organisms.

>>Think of OTUs as "bins" of similar microbes, defined by arbitrary similarity thresholds.



Happy Hacking!

Building AI **together**

ZiND!

